

1. Welcome



Building Tools To Support Urban Heat Management

This technical manual accompanies the heat-risk assessment and 3-30-300 analysis presented as part of a capstone project for the World Bank Group. The project was completed by Carnegie Mellon University students — Gabe Hafemann, Nandita Kannapadi, Nishan Sah, Sakina Mammadova, Zoë Baker — under the guidance of faculty advisor Suzy Li.

The final presentation was delivered on **December 10, 2025**.

Final Presentation link:

<https://cmu.zoom.us/rec/share/9l8tb8d4dapOd6PpxMPQ-FmDYptCJuWNj0HyJUPHBcfUHhWkd4Ox0-yU918OzVPI.MtX3EEnc0WnNcvNH>

Passcode: 5AC.KJV4

An online version of this technical manual, including interactive navigation and updates, is available at: <https://sakinamammad.github.io/Urban-Heat-Technical-Manual/>

2. Introduction

2.1. Purpose of the Codebook

The purpose of this codebook is to provide a structured framework that enables local governments and policymakers to identify priority areas for heat mitigation using spatial and satellite-based analysis. This codebook serves as both a technical guide and a documentation tool, enabling other researchers, policymakers, and local governments to understand and reproduce the team's workflow. The manual is designed to help users apply the same methodology to their own contexts, enabling them to do:

- Pinpoint areas most in need of heat mitigation actions
- Implement interventions that align with the 3-30-300 framework

2.2. Overview of the Project

Project Title

Using Satellite and GIS Data to Support Urban Heat Management

Affiliation

Carnegie Mellon University Heinz College - Capstone Project

In collaboration with the Climate Finance and Economics Unit, World Bank Group

Project Context

Urban heat is intensifying globally due to climate change, rapid urbanization, and loss of green cover, posing significant risks to health, productivity, and infrastructure. Vulnerable populations are disproportionately affected by rising temperatures, particularly in dense, low-income urban areas.

Cities require targeted, data-driven heat action plans to mitigate risk and enhance resilience. This project aims to support that need through spatial analysis and policy-relevant recommendations.

3-30-300 Framework (Cecil Konijnendijk 2021)

The 3-30-300 framework, developed by urban forester Dr. Cecil Konijnendijk, provides measurable indicators to guide equitable urban greening and cooling strategies:

- 3 trees visible from every home, school, or workplace
- 30% tree canopy cover in every neighborhood
- 300 meters or less distance from every residence to a public green space.

Pilot City: Nairobi

Nairobi, the capital of Kenya, was selected as the pilot city for this analysis due to its rapid urban growth and vulnerability to heat stress, particularly in informal settlements (Engel et al. 2025). The city's uneven distribution of vegetation and green spaces has contributed to localized urban heat islands, where surface temperatures can be several degrees higher than surrounding rural areas. These heat disparities are often correlated with income, population density, and infrastructure access, which highlights the need for data-informed, equity-centered policy planning (Shirodkar 2024).

Project Goals

The project aims to:

- Identify urban heat hotspots using open-source geospatial datasets
- Assess vulnerability and equity impacts of urban heat exposure
- Support local governments in designing data-informed, equitable heat action plans
- Provide a replicable analytical framework for future World Bank applications

Ultimately, the project's findings will help cities reduce heat-related mortality and illness, improve urban livability, and promote climate resilience.

Analytical Focus

The analysis combines climate, environmental, and socioeconomic data to assess patterns of urban heat exposure. Using the 3–30–300 framework, which examines tree visibility, canopy cover, and access to green spaces, the project evaluates the relationship between environmental factors and urban heat vulnerability.

Areas within cities are analyzed on the basis of:

- Surface urban heat island intensity
- Tree canopy and green space access
- Building density
- Social vulnerability considerations (e.g., employment, age, disability)

2.3. Targeted Stakeholders

This project is designed for individuals and organizations working in urban planning, climate change, and social equity.

Main stakeholders include:

- **City governments and urban planners:** to use data for planning green areas, zoning, and cooling strategies
- **Environmental and health agencies:** to find which neighborhoods are most affected by heat and need more attention
- **Community groups and NGOs:** to support which neighborhoods that are most affected by heat and need more attention

3. Data Requirements & Structure

2.1. Ideal & Minimum Standard Datasets

This section outlines the minimum datasets required to run the heat risk assessment, along with ideal datasets that can strengthen the analysis when available. Because the methodology is designed to be replicable across diverse cities with varying data environments, each dataset is categorized as either Required (Minimum) or Optional (Ideal).

The datasets fall into these analytical domains:

- Administrative Boundaries and Spatial Units
- Heat Exposure
- Vegetation and Green Space (3–30–300 indicators)
- Built Environment
- Social Vulnerability
- Health and Environmental Risk Indicators (Optional)

These domains reflect the project’s analytical priorities and the core components of the 3–30–300 framework.

Required and Optional Datasets

Category	Purpose	Ideal Datasets	Minimum Required Datasets	Potential Resources
Administrative Boundaries	Defines the analysis extent and demographic boundaries	High-resolution administrative micro-units	City boundary; basic administrative units (wards, districts)	National stats portals/ Generate Spatial Units
Urban Heat Exposure	Measures heat intensity	Urban Heat Island (UHI) index computed from rural–urban temperature differences	Land Surface Temperature (LST)	Landsat 8
Vegetation and Tree Canopy	Measures compliance with 30-number trees and tree canopy area	Tree-level canopy data or LiDAR vegetation structure	NDVI raster, [tree counts/survey/GSV]	NDVI

Built Environment	Captures urban form and heat absorption	Building height, materials, albedo, and impervious surface data	Building footprints	Google Open Buildings
Social Vulnerability Indicators	Proxy for density and electrification*	Local Population Survey (age, unemployment, electrification)	VIIRS mean brightness	NASA/ Local Population and Housing Survey or census
Optional Environmental Enhancements	Improves exposure characterization	Humidity, heat index, PM2.5, or surface imperviousness	None required	ERA5, WorldCover

**The indicators listed above are intended to capture social and demographic characteristics that increase individual and community-level risk under extreme heat conditions, as defined in the World Bank Group Heat Risk framework.*

Note: *Optional datasets mentioned above are provided only as suggestions to enable more advanced analyses and are not included in the technical manual instructions for implementing these elements.*

4. Methodology and Replication Instructions

This section provides step-by-step instructions to replicate the preprocessing and data integration workflow used for Nairobi. All steps are written in an instructional format so analysts can apply the same process to any new city.

Step 0: Extracting and Preparing Input Data

Extract Data

The first step is to gather all datasets required to support the heat, environmental, and social vulnerability analysis. Analysts should begin by obtaining administrative boundaries and/or creating a tessellation grid, which establishes the spatial framework for the assessment.

- Obtain **administrative boundaries** to define the study extent.
 - If administrative boundaries are unavailable, generate a **tessellation grid** using the *Generate Tessellation* tool. The size of the grid is dependent on the intervention goals of the user; however, the Nairobi pilot analysis used a size of 2 km².

- Acquire **heat exposure data**. If needed, compute UHI by subtracting rural baseline temperatures from urban LST.
- Download **vegetation data** such as NDVI
- Obtain **built environment and population data** (building footprints, population grids, land cover).
- Collect **social vulnerability indicators** such as unemployment, youth share, senior share, and disability.
- Download **nightlights rasters** as a proxy for electrification and density. (No need if social vulnerability data is available)

Ensure all datasets use compatible coordinate systems and appropriate spatial resolution before proceeding.

Clip Data

After all datasets are extracted, analysts clip each layer to the study boundary to ensure consistency across inputs. Raster datasets such as nightlights should be clipped to the boundary using the Clip Raster tool so that illumination values reflect only the area of interest. The same approach is applied to vegetation rasters such as NDVI, ensuring that greenness metrics capture only the relevant urban extent. Clipping all layers at this stage avoids later discrepancies in zonal calculations, overlays, and joins.

- If using nightlight data, clip the **nightlights raster** to the administrative boundary using *Clip Raster*.
- Clip the **NDVI raster** using the same boundary to maintain consistency.
- Clip any additional raster layers (e.g., heat rasters, land cover) to ensure uniform spatial extent.

Step 1: Measuring Heat - Urban Heat Island Effect

Note: *This step can be skipped if you have any other means of calculating the UHI effect, and have a raster ready for analysis.*

In order to calculate the Urban Heat Island Effect, a customizable toolkit has been developed. The toolkit utilizes a python script and a detailed documentation can be found in this repository:

<https://github.com/nishansrepo/SUHIIAnalysisTool>

Our measure of Urban Heat Island Effect is **Surface Urban Heat Island Intensity (SUHII)**. Surface Urban Heat Island Intensity quantifies how much hotter urban areas are compared to surrounding rural land, measured as:

$$SUHII = \mu(LST_{urban}) - \mu(LST_{rural})$$

This metric is essential for heat vulnerability assessments because it identifies where within a city the built environment amplifies thermal exposure. When combined with social vulnerability indicators (e.g., age demographics, income levels, housing quality, access to cooling), SUHII enables the construction of a composite heat risk score that pinpoints populations facing compounded risk, areas that are both physically hotter and socially less equipped to cope. SUHII-based risk mapping supports targeted interventions such as urban greening programs, cool roof initiatives, and heat-health early warning systems in the neighborhoods that need them most.

Datasets

Land Surface Temperature (LST)

- Purpose: Primary thermal input
- Recommended Source: Landsat-8/9

Land Use Land Cover (LULC)

- Purpose: Urban classification, Water pixel masking
- Recommended Source: Dynamic World (10m) (Brown et al., 2022) via Google Earth Engine

Normalized Difference Vegetation Index (NDVI)

- Purpose: Vegetation filtering
- Recommended Source: Landsat-8/9 or Sentinel-2

Digital Elevation Model

- Purpose: Elevation correction
- Recommended Source: SRTM (30m) or ASTER GDEM

Administrative Boundary

- Purpose: Study area extent
- Sources: Local government, GADM, OpenStreetMaps

How it is created

Urban Area Definition

Defines which pixels represent the 'urban' thermal signal. Three methods are available:

- **LULC-Based Method:** Urban pixels are identified using land use/land cover classification products (e.g., Dynamic World Class 6 = Built Area). This leverages validated classification products and provides consistent definitions across studies (Raj & Yun, 2024).
- **NDVI-Based Method:** Urban pixels are identified using a vegetation index threshold ($NDVI < 0.3$), based on the strong inverse correlation between LST and NDVI (Ahmad et al., 2024). Warning: This method cannot distinguish urban from bare soil or water so it should be used only when LULC data is unavailable.
- **Combined LULC + NDVI Method:** Uses LULC classification to identify built-up areas, then applies an NDVI filter to exclude vegetated pixels (parks, urban forests). This

isolates impervious surfaces which drive the UHI effect, preventing urban green spaces from biasing the urban mean downward (Zhou et al., 2014; Naidu & Chundeli, 2023).

Water Body Exclusion: All methods optionally exclude water bodies from the urban mask. Water has distinct thermal properties (high heat capacity, evaporative cooling) that would bias urban LST estimates (Mentaschi et al., 2022).

Rural Reference Selection

Defines the baseline 'rural' temperature against which urban temperatures are compared. Four methods are available:

- **Fixed Buffer Method:** Creates a fixed-width annular buffer (default: 10km) around the urban boundary. This is the most common approach, easy to replicate across studies (Raj & Yun, 2024; Zhou et al., 2014).
- **Urban Halo Method:** Skips an inner exclusion zone before the buffer begins, avoiding thermal contamination from the urban heat island 'footprint' that extends beyond administrative boundaries through heat advection (Mentaschi et al., 2022). Recommended for large cities with 1-3km inner exclusion.
- **Three-Ring Method:** Creates dynamically-sized buffer zones scaled to city area using formulas from (Fernandes et al., 2024). Three concentric zones are defined: Urban Adjacent (Ua), Future Urban Adjacent (FUa), and Peri-Urban (PUa). This enables consistent cross-city comparisons regardless of city size.
- **In-City Method:** Selects rural reference pixels from within the administrative boundary itself (parks, urban forests, undeveloped land). Useful for intra-urban analysis or isolated cities where the surrounding landscape is climatically dissimilar. Additionally calculates Normalized UHI following (Ahmad et al., 2024): $(\mu(LST_{urban}) - \mu(LST_{rural})) / \sigma$, enabling cross-city comparison regardless of absolute temperatures.

Rural Pixel Filtering

Regardless of the geometry method, rural pixels undergo additional filtering to ensure climatically comparable vegetated land. There are following options available:

- **Vegetation (NDVI):** Set above a certain threshold. Ensures rural pixels are vegetated, not bare soil or degraded land.
- **Water Exclusion:** Water bodies have high thermal inertia (~4x soil) and evaporative cooling, which would artificially depress rural LST and inflate SUHI (Raj & Yun, 2024; Mentaschi et al., 2022).
- **Urban LULC Exclusion:** Removes any built-up pixels that fall within the buffer zone.
- **NoData Exclusion:** Removes pixels with invalid LULC values (e.g., class 255).
- **Distance Filter:** Optional minimum distance from urban edge, excluding pixels in the UHI 'footprint' zone. Uses Euclidean Distance Transform.
- **Elevation Filter:** The environmental lapse rate (~6.5C/km) causes systematic temperature differences with elevation. Adaptive filtering starts at +/-50m and expands if needed (Raj & Yun, 2024).

SUHII Calculation

The primary output is the scalar SUHII value representing the absolute temperature difference between mean urban LST and mean filtered rural LST, reported in degrees Celsius. This formulation is used by (Raj & Yun, 2024) and (Fernandes et al., 2024).

For the in-city method, an additional Normalized UHI metric is calculated following (Ahmad et al., (2024), expressing the UHI as standard deviations from the study area mean, enabling cross-city comparison regardless of absolute temperatures.

Output

The primary output is a pixel-level deviation raster covering the full urban boundary. Each pixel value represents how much warmer (positive) or cooler (negative) that location is relative to the rural baseline:

$$Deviation_{pixel} = LST_{pixel} - \mu(LST_{rural})$$

The raster is masked to the urban administrative boundary. The final filtered rural mask (after vegetation, water, LULC, and elevation filtering) is used to compute the baseline rural mean temperature.

Aggregating to Spatial Units

- In your GIS software, add the SUHII raster output and the vector spatial units as layers.
- Calculate the zonal statistics (mean of raster pixels in each spatial unit) and join it to your vector layer.

This spatially-explicit output can be processed with social vulnerability indices to generate neighborhood-level heat risk scores for planning and intervention prioritization.

Step 2: Social Vulnerability

To identify populations most at risk under extreme heat, create a **Social Vulnerability Score** for each spatial unit. The goal of this step is to capture demographic and socioeconomic characteristics that influence how different communities experience and recover from exposure to heat. Because Nairobi's population census data is published using relatively large enumeration boundaries, the pilot analysis combined census indicators with nightlight and building-density data to approximate sub-neighborhood variation (Andries et al. 2023). The efficacy has not been 100% confirmed at the local level, so we strongly encourage cities to use whatever local data they may have.

Census indicators are assigned to tessellations based on the sub-county they overlapped the most. These layers together provide a more complete picture of vulnerability across the city.

How to create it

Generate the Social Vulnerability Score using a Suitability Analysis Tool, where each field represents a variable associated with social vulnerability. For our analysis, these variables included:

- Unemployment (positive relationship to vulnerability)
- % Youth (positive)
- % Seniors (positive)
- % Disabled (positive)
- Mean electrification (inverse relationship — lower electrification corresponds to higher vulnerability)

***Note:** If planners are using QGIS instead of ArcGIS Pro, this process can be replicated through a multi-criteria overlay, or by manually standardizing inputs through a calculated field.*

Standardizing and Weighting Inputs

All measures are standardized so they can be compared on the same scale.

Weights can be set at the discretion of planners and may vary depending on local priorities.

However, unless there is a clear rationale for prioritizing specific demographic groups, we recommend equal weighting for all indicators.

For the analysis, weigh all vulnerability variables equally, with the exception of building density, which is weighted lower at **8%**, while all other variables remain at **18.4%** each.

Combining Indicators

If using a suitability analysis approach, the Suitability Analysis Tool standardizes the input variables and applies user-defined weights to produce a final **Social Vulnerability Score** for each tessellation.

Output: Social Vulnerability Index

The resulting score is used as a **Social Vulnerability Index (SVI)** that can be applied to compare relative vulnerability across spatial units within the city. The SVI can be classified into multiple levels (e.g., low to high vulnerability) to support visualization, grouping of areas by risk, and integration with other analytical components, such as heat exposure metrics.

Step 3: Calculating Holistic Heat Risk

After generating the SUHII layer (heat exposure) and the Social Vulnerability Index (SVI), the next step is to combine these two components into a single Holistic Heat Risk Score. This score captures both the physical intensity of heat and the degree to which local populations are

socially or economically more sensitive to heat impacts. Combining these layers allows planners to identify locations where extreme temperatures intersect with high vulnerability, ensuring interventions are directed where they are most needed.

The approach follows the logic used in tree equity and heat-justice frameworks, where exposure and vulnerability together define relative risk. Because both layers were already normalized and summarized to the same 2 km² tessellations, they can be combined directly.

How it is created

The following formula is used to calculate Holistic Heat Risk for each spatial unit:

$$\text{Urban Heat Risk} = \text{SUHII} \times \text{Social Vulnerability Index} \times 100$$

This multiplicative approach ensures that areas with both high heat exposure and high vulnerability score the highest on the risk scale. If either exposure or vulnerability is low, the overall risk is moderated, aligning with the goal of identifying neighborhoods where both environmental and social burdens are elevated.

Output: Holistic Heat Risk Map

The final result is a heat risk map that highlights priority zones for intervention. This layer serves as the bridge between Steps 1 and 2 and the subsequent 3-30-300 assessment, ensuring that greening and cooling strategies can be compared with underlying heat risk.

Step 4: Identifying Areas Needing 3-30-300 Improvement

The final component of the analysis involves assessing each spatial unit against the 3-30-300 framework, which evaluates access to trees and green space through three indicators:

- Visibility of at least 3 trees from buildings,
- 30% tree canopy cover, and
- 300 meters access to a public park.

The goal of this step is to understand where green-infrastructure gaps align with high heat exposure and high social vulnerability. By calculating each indicator at the tessellation level, it is possible to generate a consistent greenness profile for the entire city and identify areas that fall short of meeting the 3-30-300 standards (Browning et al. 2024).

Three Trees Within View

This indicator evaluates tree visibility from buildings by assessing whether a minimum number of trees are present within a defined viewshed around each structure (Liang et al. 2023). The metric is intended to capture residents' visual exposure to trees rather than overall canopy coverage. Ideally, cities will have access to point-level locations of individual trees. In this case, tree visibility can be assessed directly by counting how many individual trees fall within each

building's viewshed buffer. If individual tree locations are not available, tree visibility can be approximated using buffer-based methods derived from vegetation or proxy datasets. In these cases, buffers may be applied to represent tree presence and visibility within a reasonable distance from buildings, based on available data and local context.

How it is created

- Generate building-level viewshed buffers to approximate the area from which trees may be visible.
- If individual tree point data are available:
 - Perform a spatial join to count the number of trees within each building's viewshed buffer.
- If individual tree point data are not available:
 - Use buffered vegetation features or proxy tree layers to approximate tree presence within the viewshed area.
- Classify buildings based on tree visibility:
 - no data
 - fewer than three trees
 - at least three trees
- Aggregate building-level results to tessellations by calculating the percentage of buildings meeting the three-tree threshold within each spatial unit.

Thirty Percent Tree Canopy Cover

To assess canopy coverage, use NDVI from the summer season with <10% cloud cover. Applying an NDVI threshold allowed us to identify areas of tree canopy within each tessellation.

How it is created

- Filter NDVI pixels using a >0.5 threshold for tree canopy
- Calculate the canopy area within each tessellation
- Compute percent canopy cover per unit
- Classify each tessellation as either meeting or not meeting the 30% canopy goal

Three Hundred Meters to a Park

To evaluate park access, digitize public parks within the city limits and measure building-level access to these green spaces.

How it is created

- Digitize parks inside the study area
- Perform a 300 m near-table analysis to determine which buildings were within 300 meters of a park*
- Classify buildings as inside or outside the 300 m boundary

- Summarize results to tessellations by calculating the percentage of buildings within 300 m

**QGIS users can use the NNJoin PlugIn: <https://plugins.qgis.org/plugins/NNJoin/>*

Greenness Index

Once the three indicators were calculated, combine them into a single greenness score to summarize overall compliance with the 3-30-300 framework.

How it is created

- Equally weigh the three components:
 - tree visibility
 - canopy cover
 - park access
- Produce a 0–1 normalized greenness index for each tessellation

This greenness index allows direct comparison of green-infrastructure gaps across the city and serves as an essential input to the final priority mapping.

Priority Areas for Intervention

Finally, combine the greenness index with the Holistic Heat-Risk score to identify areas where high social vulnerability and low greenness levels overlap. These locations represent the highest-priority sites for heat-mitigating interventions such as tree planting, canopy restoration, and expanded park access.

How it is created

- Combine the Social Vulnerability Index and greenness results
- Classify tessellations where SVI is high and the green score is low
- Flag these as priority intervention zones

Dashboard Development (Optional for Analysis)

The dashboard is intended as a communication tool to help advocates and practitioners present results to policymakers and other non-technical audiences who may not have a detailed understanding of SUHI/SUHII or the 3-30-300 framework. While the dashboard is not required to complete the analysis, it provides a clear and accessible way to visualize results and support decision-making. Users may choose to reproduce the dashboard if desired.

The dashboard is designed to produce a series of map layouts that show where areas of high priority overlap with locations targeted for tree planting and park construction. Presenting results in this format can yield insights similar to static map layouts while requiring less time and effort to update or explore.

Foundational Analysis in a Web Map

To build the dashboard, users must first create a web map, either directly in ArcGIS Online or by uploading a map constructed in ArcGIS Pro. The web map should include the following layers:

- The SUHII × Social Vulnerability analysis, visualized using quantiles and a red color ramp, where lighter shades represent lower scores and darker shades represent higher scores.
- A visualization of the Greenness Score, as described earlier in the manual.
- A layer highlighting areas of noncompliance with the “300” element of the framework, visualized semi-transparently.
- A layer highlighting areas of noncompliance with the “3–30” elements of the framework, visualized semi-transparently and distinctly from the “300” noncompliance layer.
- A priority sketch layer highlighting areas of overlap. Pop-ups for these sketches should be edited to include comments on findings and potential interventions.

This web map serves as the foundational element of the dashboard.

Expanding the Dashboard

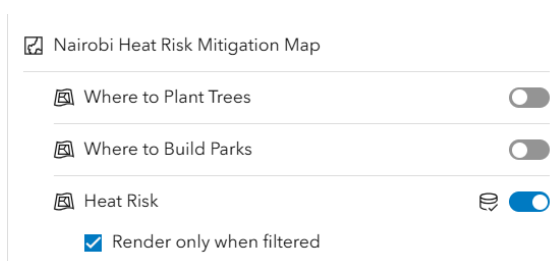
Once the layers have been uploaded into a web map, create a new dashboard draft in ArcGIS Online and add the web map. Add a title bar, a legend, and two sidebars. Text from the Nairobi Demonstration document may be copied and edited as needed to support visualization and interpretation.

Filters

To enable interactive exploration, filters can be added to the dashboard.

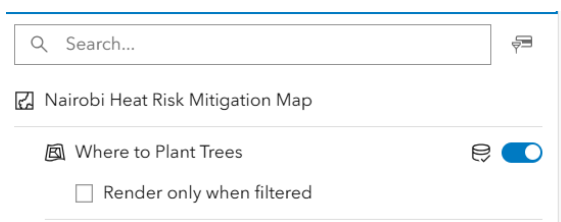
Heat Risk Filter

- Add a Number Selector element to the title bar.
- Set the display type to Combination with a Range input type.
- Enable the Show Reset option for ease of use.
- Under Limits from, select Statistic.
- Choose the Heat Risk layer, and for the Field, use the combined Social Vulnerability and SUHII value created during the analysis.
- Set Precision to 1 and Increment to 1.
- Under Actions, configure the filter so the Heat Risk layer is active and rendered only when filtered.



Plant Trees Filter

- Add a second Number Selector element to the title bar.
- Use a Combination display with Range inputs and enable the reset option.
- Under Limits from, select Statistic.
- Choose the Where to Plant Trees layer.
- For the filter condition, use “Plant Trees” is equal to “Plant Trees in this Area,” as defined in the previous analyses.
- For the Field, select the tree canopy cover percentage variable.
- Set Precision to 1 and Increment to 1.
- Under Actions, confirm that the filter configuration matches the intended screen layout.



Using the Dashboard

These elements, in combination, allow you to identify areas ripe for intervention.

5. Limitations and Future Work

Limitations

While the workflow developed for this project provides a replicable approach to assessing urban heat risk and evaluating city-level performance against the 3-30-300 framework, several limitations should be acknowledged. These limitations reflect constraints in data availability, methodological assumptions, and the absence of local validation at this stage.

Tree Data Availability

The accuracy of the “three trees within view” and canopy metrics depends heavily on the quality and completeness of available tree data. In many cities, including Nairobi, publicly accessible datasets may be incomplete or outdated, which can lead to underestimation or misclassification of green cover.

Park Definitions and Calculations

Park boundaries were digitized manually for this pilot analysis. As park quality, accessibility, and official boundaries vary across cities, results may differ depending on how parks are defined or which areas are included. These methodological differences can affect the 300-meter access metric.

Placement of Interventions

The analysis identifies priority tessellations but does not determine the optimal placement of specific interventions (e.g., where exactly to plant trees within a hexagon). More advanced siting tools or optimization models would be required to translate priority areas into project-ready intervention plans.

QGIS Integration

Although the workflow was executed in ArcGIS Pro, full integration into QGIS has not yet been completed. Some processes such as suitability analysis or near-table generation may require alternate tools or manual adaptations in QGIS environments.

Future Work

Several opportunities exist to expand and strengthen this analytical framework as additional data, tools, and local expertise become available. The following areas outline future work that can improve the accuracy, usability, and policy relevance of the heat-risk and 3-30-300 assessment:

Deep Learning for Tree Crown Identification

Future iterations of this project could incorporate deep learning methods to automatically detect individual tree crowns from high-resolution imagery. This would improve the accuracy of the “three trees within view” metric and reduce reliance on external datasets.

Optimization of Intervention Placement

While this analysis identifies priority tessellations, further work is needed to determine the optimal locations for specific interventions (e.g., where exactly to plant trees or expand canopy). Linear programming or other optimization tools could support site-level decision-making.

QGIS Plugin Development

Developing a QGIS plugin to automate components of the workflow, including tessellation creation, zonal statistics, and SUHI filtering would make the methodology more accessible for cities without ArcGIS licenses.

Validation with Local Planners

Collaboration with Nairobi city planners and technical teams is essential to refine assumptions, verify feasibility, and ensure the results align with local land-use contexts. Field validation would strengthen the credibility and usability of the final outputs.

Integration with Broader Climate-Risk Tools

Future versions of this workflow could integrate urban heat metrics and 3-30-300 results with the World Bank’s broader climate-risk and urban resilience platforms, enabling a more comprehensive, hyper-local understanding of vulnerability across multiple hazards.

6. References and Data Licensing

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Data references

Dataset	Source	URL	License
Administrative Boundaries	GADM	https://gadm.org/data.html	Open Data
LST	Landsat-8/9, USGS / NASA	https://earthexplorer.usgs.gov/	Public Domain
NDVI	Sentinel-2, Copernicus (ESA)	https://browser.dataspace.copernicus.eu/	Public Domain
LULC	Dynamic World, Google Earth Engine	https://developers.google.com/earth-engine/datasets/catalog/GOOGLE_DYNAMICWORLD_V1	Public Domain (Requires a free GEE license)
Elevation	SRTM 30m, USGS / NASA	https://earthexplorer.usgs.gov/	Public Domain
VIIRS Nighttime Lights (VNP46A4)	NASA LP DAAC	https://ladsweb.modaps.eosdis.nasa.gov/missions-and-measurements/products/VNP46A4/#overview	Public Domain

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